

RHCSA EXAM**3 Hrs**

You have to configure servera.lab.example.com and serverb.lab.example.com with the following requirements.

Your Domain name:- lab.example.com

Your Network:- 172.25.250.0/255.255.255.0

Hostname:- servera.lab.example.com and serverb.lab.example.com

Your root user password for serverb.lab.example.com is **redhat**

Your ip address for servera.lab.example.com is **172.25.250.10**

Container registry server is **utility.lab.example.com**

Use **admin** as username and **redhat** as password for container registry

serverb.lab.example.com

1. Configure the network

a) Assign Hostname and Ip address for your virtual machine.

Hostname serverb.lab.example.com

IP Address 172.25.250.11

Netmask 255.255.255.0

Gateway 172.25.250.254

Nameserver 172.25.250.254

```
root@RH-02: # hostnamectl set-hostname serverb.lab.example.com
root@RH-02: # nmcli connection modify enp1s0 ipv4.addresses 172.25.250.11/24
root@RH-02: # nmcli connection modify enp1s0 ipv4.gateway 172.25.250.254
root@RH-02: # nmcli connection modify enp1s0 ipv4.dns 172.25.250.254
root@RH-02: # nmcli connection modify enp1s0 ipv4.method manual
root@RH-02: # nmcli connection down enp1s0
Connection 'enp1s0' successfully deactivated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/2)
root@RH-02: # nmcli connection up enp1s0
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/3)
root@RH-02: # hostname
serverb.lab.example.com
root@RH-02: # ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp1s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 52:54:00:a0:55:18 brd ff:ff:ff:ff:ff:ff
    altname enx525400a05518
    inet 172.25.250.11/24 brd 172.25.250.255 scope global noprefixroute enp1s0
        valid_lft forever preferred_lft forever
    inet6 fe80::5054:ff:fea0:5518/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
root@RH-02: # ip route
default via 172.25.250.254 dev enp1s0 proto static metric 100
172.25.250.0/24 dev enp1s0 proto kernel scope link src 172.25.250.11 metric 100
root@RH-02: #
```

2. Create a repository file

http://classroom.example.com/content/rhel9.0/x86_64/dvd/AppStream

http://classroom.example.com/content/rhel9.0/x86_64/dvd/BaseOS

```

[local-BaseOS]
name=BaseOS
baseurl=http://classroom.example.com/content/rhel9.0/x86_64/dvd/BaseOS
enabled=1
gpgcheck=0

[local-AppStream]
name=AppStream
baseurl=http://classroom.example.com/content/rhel9.0/x86_64/dvd/AppStream
enabled=1
gpgcheck=0

```

3. Configure the Selinux

The webserver is running on non-standard port 82 having a issue serve the web content. Debug and fix the issue:

- The webserver can serve all the existing HTML file located at /var/www/html directory (Don't alter or remove any files in this directory)
- The webserver can serve the content on port 82.
- Make the content accessible.

```

Complete!
root@RH-02:~#
root@RH-02:~# semanage port -a -t http_port_t -p tcp 82
root@RH-02:~# firewall-cmd --permanent --add-port=82/tcp
success
root@RH-02:~# firewall-cmd --reload
success
root@RH-02:~# restorecon -Rv /var/www/html
root@RH-02:~# systemctl restart httpd
root@RH-02:~# semanage port -l | grep http_port_t
http_port_t                tcp      82, 80, 81, 443, 488, 8008, 8009, 8443, 9000
pegasus_http_port_t        tcp      5988
root@RH-02:~#

```

4. Create the following users, groups and group memberships:

- a) A group named admin.
- b) A user harry who belongs to admin as a secondary group.
- c) A user natasha who belongs to admin as a secondary group.
- d) A user sarah who does not have access to an interactive shell on the system and who is not member of admin.
- e) harry, natasha and sarah should have password of password.

```
root@RH-02:~# groupadd admin
groupadd: group 'admin' already exists
root@RH-02:~# useradd -G admin harry
useradd: user 'harry' already exists
root@RH-02:~# useradd -G admin natasha
useradd: user 'natasha' already exists
root@RH-02:~# useradd -s /sbin/nologin sarah
useradd: user 'sarah' already exists
root@RH-02:~# echo 'harry:password' | chpasswd
root@RH-02:~# echo 'natasha:password' | chpasswd
root@RH-02:~# echo 'sarah:password' | chpasswd
root@RH-02:~# id harry
uid=1001(harry) gid=1002(harry) groups=1002(harry),1001(admin)
root@RH-02:~# id natasha
uid=1002(natasha) gid=1003(natasha) groups=1003(natasha),1001(admin)
root@RH-02:~# id sarah
uid=1003(sarah) gid=1004(sarah) groups=1004(sarah)
root@RH-02:~# tail -n 3 /etc/passwd
harry:x:1001:1002::/home/harry:/bin/bash
natasha:x:1002:1003::/home/natasha:/bin/bash
sarah:x:1003:1004::/home/sarah:/sbin/nologin
root@RH-02:~# tail -n 3 /etc/shadow
harry:$y$j9T$Q1.H0ZF8z.xr0nwBHgXT/$ISH1cx6nFxtMo1wiTPp/dmRrgfcbjKtDRt5QGKR3Y09:20673:0:99999:7:::
natasha:$y$j9T$xzE7NAX7CMA3rzgASH3ut0$pax7kwNSjZui8iJfi/krt6s8j7WeQe5sDRh5WDLFrP.:20673:0:99999:7:::
sarah:$y$j9T$Q13P0mzuWaDOGA8A1blHl0$.2ms0F6UDerCQWRinCRh4JWHubsnP4wGcnNCzxKw9R1:20673:0:99999:7:::
root@RH-02:~#
```

5. Create a collaborative directory /common/admin with the following characteristics:

- a) Group ownership of /common/admin is admin.
- b) The directory should be readable, writable and accessible to members of admin, but not to any other user.
(It is understood that root has access to all files and directories on the system.)
- c) Files created in /common/admin automatically have group ownership set to the admin group.

```
root@RH-02:~# mkdir -p /common/admin
root@RH-02:~# chown :admin /common/admin
root@RH-02:~# chmod 2770 /common/admin
root@RH-02:~# ls -ld /common/admin
drwxrws---. 2 root admin 6 Aug  8 05:48 /common/admin
root@RH-02:~#
```

6. Configure autofs to automount the home directories of production5 domain users. Note the following:

- a) servera.lab.example.com (172.25.250.10) NFS -exports /user-homes to your system.
- b) production5 home directory is servera.lab.example.com:/user-homes/production5
- c) production5 home directory should be automounted locally beneath to /localhome as /localhome/production5 .
- d) home directories must be writable by their users.
- e) production5's password is redhat.
- f) While you are able to log in as any of the users production1 through production30, the only home directory that is accessible from your system is production5.

```

root@servera:~# hostname
servera.lab.example.com
root@servera:~# systemctl enable --now nfs-server
mkdir -p /user-homes/production5
chmod -R 777 /user-homes
chown -R root:root /user-homes
echo "/user-homes *(rw, sync, no_root_squash)" > /etc/exports
exportfs -rva
firewall-cmd --permanent --add-service=nfs
firewall-cmd --permanent --add-service=rpc-bind
firewall-cmd --permanent --add-service=mountd
firewall-cmd --reload

setsebool -P nfs_export_all_rw 1
setsebool -P nfs_export_all_ro 1
exporting */user-homes
Warning: ALREADY_ENABLED: nfs
success
success
success
success
root@servera:~#

```

```

root@serverb:~# hostname
serverb.lab.example.com
root@serverb:~#
useradd -u 2005 -d /localhome/production5 production5
echo "/localhome /etc/auto.localhome" > /etc/auto.master.d/localhome.autofs
echo "* -rw, sync servera.lab.example.com:/user-homes/&" > /etc/auto.localhome
systemctl daemon-reload
systemctl restart autofs
systemctl enable autofs
useradd: user 'production5' already exists

```

7. The user harry must configure cron job that runs daily at 12:30 local time and execute/bin/echo "hello".

```
root@RH-02:~# crontab -u harry -e
no crontab for harry - using an empty one
crontab: installing new crontab
root@RH-02:~# crontab -u harry -l
30      12      *      *      *      /bin/echo "hello"
root@RH-02:~#
```

8. Configure the NTP

- a) Configure your system so that it is an NTP client of classroom.example.com

```
root@RH-02:~# sed -i '1i server classroom.example.com iburst' /etc/chrony.conf
root@RH-02:~# systemctl restart chronyd
root@RH-02:~# chronyc makestep
200 OK
root@RH-02:~#
```

9. Locate the Files

- a) Locate all the files owned by sarah and make a copy of them in the given path /root/find.user

```

root@RH-02:~# mkdir -p /root/find.user
root@RH-02:~# find / -user sarah -exec cp -a {} /root/find.user/ \;
find: '/proc/6278/task/6278/fd/5': No such file or directory
find: '/proc/6278/task/6278/fdinfo/5': No such file or directory
find: '/proc/6278/fd/6': No such file or directory
find: '/proc/6278/fdinfo/6': No such file or directory
cp: '/root/find.user/sarah' and '/root/find.user/sarah' are the same file
cp: '/root/find.user/.mozilla' and '/root/find.user/.mozilla' are the same file
cp: '/root/find.user/extensions' and '/root/find.user/extensions' are the same file
cp: '/root/find.user/plugins' and '/root/find.user/plugins' are the same file
cp: '/root/find.user/.bash_logout' and '/root/find.user/.bash_logout' are the same file
cp: '/root/find.user/.bash_profile' and '/root/find.user/.bash_profile' are the same file
cp: '/root/find.user/.bashrc' and '/root/find.user/.bashrc' are the same file
cp: cannot overwrite directory '/root/find.user/sarah' with non-directory '/var/spool/mail/sarah'
root@RH-02:~# ls -la /root/find.user
total 16
drwxr-xr-x. 6 root root 124 Aug 8 06:06 .
dr-xr-x---. 17 root root 4096 Aug 8 06:05 ..
-rw-r--r--. 1 sarah sarah 18 Oct 28 2024 .bash_logout
-rw-r--r--. 1 sarah sarah 144 Oct 28 2024 .bash_profile
-rw-r--r--. 1 sarah sarah 522 Oct 28 2024 .bashrc
drwxr-xr-x. 2 sarah sarah 6 Oct 28 2024 extensions
drwxr-xr-x. 4 sarah sarah 39 Jul 22 15:20 .mozilla
drwxr-xr-x. 2 sarah sarah 6 Oct 28 2024 plugins
drwx-----. 3 sarah sarah 78 Aug 8 05:40 sarah
root@RH-02:~#

```

10. Find the string

a) Find a string "home" in /etc/passwd and searching string as been stored in /root/search.txt

```

root@RH-02:~# grep "home" /etc/passwd > /root/search.txt
root@RH-02:~# cat /r
root/ run/
root@RH-02:~# cat /root/search.txt
sirrootfire:x:1000:1000:sirrootfire:/home/sirrootfire:/bin/bash
harry:x:1001:1002::/home/harry:/bin/bash
natasha:x:1002:1003::/home/natasha:/bin/bash
sarah:x:1003:1004::/home/sarah:/sbin/nologin
root@RH-02:~#

```

11. Create an user account

a) Create an user account with Userid 1326 and user name as alies.

```

root@RH-02:~# useradd -u 1326 alies
root@RH-02:~# id alies
uid=1326(alies) gid=1326(alies) groups=1326(alies)
root@RH-02:~#

```

12. Create a tar archive file

a) Backup the /var/tmp as /root/test.tar.gz

```
root@RH-02:~# tar -czvf /root/test.tar.gz /var/tmp
tar: Removing leading `/' from member names
/var/tmp/
/var/tmp/systemd-private-95af0c34653b44c1a0513e031f6aa1ed-dbus-broker.service-A3oYe7/
/var/tmp/systemd-private-95af0c34653b44c1a0513e031f6aa1ed-dbus-broker.service-A3oYe7/tmp/
/var/tmp/systemd-private-95af0c34653b44c1a0513e031f6aa1ed-irqbalance.service-Jhup8p/
/var/tmp/systemd-private-95af0c34653b44c1a0513e031f6aa1ed-irqbalance.service-Jhup8p/tmp/
/var/tmp/systemd-private-95af0c34653b44c1a0513e031f6aa1ed-polkit.service-sLfJ20/
/var/tmp/systemd-private-95af0c34653b44c1a0513e031f6aa1ed-polkit.service-sLfJ20/tmp/
/var/tmp/systemd-private-95af0c34653b44c1a0513e031f6aa1ed-rtkit-daemon.service-DfvPOP/
/var/tmp/systemd-private-95af0c34653b44c1a0513e031f6aa1ed-rtkit-daemon.service-DfvPOP/tmp/
/var/tmp/systemd-private-95af0c34653b44c1a0513e031f6aa1ed-switcheroo-control.service-nWnESd/
/var/tmp/systemd-private-95af0c34653b44c1a0513e031f6aa1ed-switcheroo-control.service-nWnESd/tmp/
/var/tmp/systemd-private-95af0c34653b44c1a0513e031f6aa1ed-systemd-logind.service-nzu9Bk/
/var/tmp/systemd-private-95af0c34653b44c1a0513e031f6aa1ed-systemd-logind.service-nzu9Bk/tmp/
/var/tmp/systemd-private-95af0c34653b44c1a0513e031f6aa1ed-upower.service-yK9dH1/
/var/tmp/systemd-private-95af0c34653b44c1a0513e031f6aa1ed-upower.service-yK9dH1/tmp/
/var/tmp/systemd-private-95af0c34653b44c1a0513e031f6aa1ed-ModemManager.service-ORxkeys/
/var/tmp/systemd-private-95af0c34653b44c1a0513e031f6aa1ed-ModemManager.service-ORxkeys/tmp/
/var/tmp/systemd-private-95af0c34653b44c1a0513e031f6aa1ed-colord.service-xSWClE/
/var/tmp/systemd-private-95af0c34653b44c1a0513e031f6aa1ed-colord.service-xSWClE/tmp/
/var/tmp/systemd-private-95af0c34653b44c1a0513e031f6aa1ed-spice-vdagentd.service-0rAhLx/
/var/tmp/systemd-private-95af0c34653b44c1a0513e031f6aa1ed-spice-vdagentd.service-0rAhLx/tmp/
/var/tmp/systemd-private-95af0c34653b44c1a0513e031f6aa1ed-fwupd.service-UIJyX1/
/var/tmp/systemd-private-95af0c34653b44c1a0513e031f6aa1ed-fwupd.service-UIJyX1/tmp/
/var/tmp/systemd-private-95af0c34653b44c1a0513e031f6aa1ed-httpd.service-W21kL5/
/var/tmp/systemd-private-95af0c34653b44c1a0513e031f6aa1ed-httpd.service-W21kL5/tmp/
/var/tmp/systemd-private-95af0c34653b44c1a0513e031f6aa1ed-chrond.service-GFBPAQ/
/var/tmp/systemd-private-95af0c34653b44c1a0513e031f6aa1ed-chrond.service-GFBPAQ/tmp/
root@RH-02:~# ls -lh /root/test.tar.gz
-rw-r--r--. 1 root root 725 Aug  8 06:11 /root/test.tar.gz
root@RH-02:~#
```

13. Build a container as student user.

a) Using the URL to build the container image with name monitor.

<http://classroom.example.com/Containerfile>

c) Do not modify the container file

14. Configure the container as a system start-up service and mount volumes persistently

a) Create the container name as ascii2pdf as student user

b) Run the container by using image monitor which one was already done in previous

c) Create the container as a system start-up service, While reboot it will automatically start the service without any human intervention.

- d) The system service should be container-ascii2pdf.
- e) The local directory /opt/files should be persistently mount on container's /opt/incomingdirectory.
- f) The local directory /opt/processed should be persistently mount on container's /opt/outgoingdirectory.

Note: In working of service starts, any file create/store under the /opt/files automatically creates into pdf on /opt/outgoing directory.

```

student@serverb:~/config/systemd/user -- bash
root@serverb: # mkdir -p /opt/files /opt/processed
root@serverb: # chmod 777 /opt/files /opt/processed
root@serverb: # chcon -R -t container_file_t /opt/files /opt/processed
root@serverb: # su - student
Last login: Sat Aug 8 12:16:00 EDT 2026 on pts/0
student@serverb: $ podman build -t monitor http://classroom.example.com/Containerfile
Error: prepping temporary context directory: Get "http://classroom.example.com/Containerfile": dial tcp 172.25.250.254:80: connect: no route to host
student@serverb: $ podman run -d --name ascii2pdf \
-v /opt/files:/opt/incoming:Z \
-v /opt/processed:/opt/outgoing:Z \
monitor
Error: ^C
student@serverb: $ mkdir -p ~/.config/systemd/user/
student@serverb: $ cd ~/.config/systemd/user/
student@serverb: ~/.config/systemd/user$ podman generate systemd --name ascii2pdf --files --new
DEPRECATED command:
It is recommended to use Quadlets for running containers and pods under systemd.

Please refer to podman-systemd.unit(5) for details.
Error: ascii2pdf does not refer to a container or pod: no pod with name or ID ascii2pdf found: no such pod: no container with name or ID "ascii2pdf" found: no such containe
r
student@serverb: ~/.config/systemd/user$ systemctl --user daemon-reload
systemctl --user enable --now container-ascii2pdf.service
Failed to enable unit: Unit container-ascii2pdf.service does not exist
student@serverb: ~/.config/systemd/user$

```

15.1 Set the Permission

- a) All new creating files for user natasha as -r----- as default permission.
- b) All new creating directories for user natasha as dr-x----- as default permission.

```

root@RH-02:~# echo "umask 0277" >> /home/natasha/.bashrc
root@RH-02:~# su - natasha
Last login: Sat Aug 8 05:47:31 EDT 2026 on pts/0
natasha@serverb:~$ touch /tmp/testfile_natasha && mkdir /tmp/testdir_natasha
natasha@serverb:~$ ls -ld /tmp/testfile_natasha /tmp/testdir_natasha
dr-x-----. 2 natasha natasha 6 Aug 8 06:52 /tmp/testdir_natasha
-r-----. 1 natasha natasha 0 Aug 8 06:52 /tmp/testfile_natasha
natasha@serverb:~$

```

15.2 Set the Password expire date

- a) The password for all new users in serverb.lab.example.com should expires after 20 days.

```

root@RH-02:~# sed -i 's/^PASS_MAX_DAYS.*PASS_MAX_DAYS 20/' /etc/login.defs
root@RH-02:~# grep "^PASS_MAX_DAYS" /etc/login.defs
PASS_MAX_DAYS 20
root@RH-02:~#

```

15.3 Assign Sudo Privilege

Assign the Sudo Privilege for Group "admin" and Group members can administrate without any password.

```
root@RH-02:~# echo "%admin ALL=(ALL) NOPASSWD: ALL" > /etc/sudoers.d/admin
root@RH-02:~# chmod 0440 /etc/sudoers.d/admin
root@RH-02:~# cat /etc/sudoers.d/admin
%admin ALL=(ALL) NOPASSWD: ALL
root@RH-02:~# su - harry
harry@serverb:~$ sudo -l
Matching Defaults entries for harry on serverb:
    !visiblepw, always_set_home, match_group_by_gid, always_query_group_plugin, env_reset, env_keep="COLORS DISPLAY HOSTNAME HISTSIZE KDEDIR LS_COLORS", env_keep+="MAIL
    PS1 PS2 QDIR USERNAME LANG LC_ADDRESS LC_CTYPE", env_keep+="LC_COLLATE LC_IDENTIFICATION LC_MEASUREMENT LC_MESSAGES", env_keep+="LC_MONETARY LC_NAME LC_NUMERIC
    LC_PAPER LC_TELEPHONE", env_keep+="LC_TIME LC_ALL LANGUAGE LINGUAS _XKB_CHARSET XAUTHORITY", secure_path=/sbin\:/bin\:/usr/sbin\:/usr/bin

User harry may run the following commands on serverb:
    (ALL) NOPASSWD: ALL
harry@serverb:~$
```

15.4 Configure the application RHCSA as an alies user, When login it will show the message "Welcome to Advantage Pro"

```
root@RH-02:~# echo 'echo "Welcome to Advantage Pro"' >> /home/alies/.bashrc
root@RH-02:~# chown alies:alies /home/alies/.bashrc
root@RH-02:~# su - alies
Welcome to Advantage Pro
alies@serverb:~$
```

15.5 Create the script file

- Create a mysearch script file under /usr/local/bin to locate files under /usr/share directory having size less than 1M.
- After executing the mysearch script file and listed(searched) files has to be copied under /root/myfiles.

```
root@RH-02:~# cat << 'EOF' > /usr/local/bin/mysearch
#!/bin/bash
mkdir -p /root/myfiles
find /usr/share -type f -size -1M -exec cp -a {} /root/myfiles/ \; 2>/dev/null
EOF
root@RH-02:~# chmod +x /usr/local/bin/mysearch
root@RH-02:~# /usr/local/bin/mysearch
root@RH-02:~# ls -la /root/myfiles | head -n 15
total 8
drwxr-xr-x.  2 root root 4096 Aug  8 07:03 .
dr-xr-x---. 18 root root 4096 Aug  8 07:03 ..
-rw-r--r--.  1 root root   0 Oct 28  2024 ca-bundle.legacy.default.crt
-rw-r--r--.  1 root root   0 Oct 28  2024 ca-bundle.legacy.disable.crt
-rw-r--r--.  1 root root   0 Dec  4  2024 Compose
-rw-r--r--.  1 root root   0 Nov 12  2024 .gitignore
-r--r--r--.  1 root root   0 Jan  6  2025 help-trailer.tmpl
-rw-r--r--.  1 root root   0 Jul 22 15:32 icons
-rw-r--r--.  1 root root   0 Oct 28  2024 iso_3166-3.xml
-rw-r--r--.  1 root root   0 Jul 28  2024 XI18N_OBJS
-rw-r--r--.  1 root root   0 Dec  4  2024 XLC_LOCALE
root@RH-02:~#
```

servera.lab.example.com

1. Assign root user password as northate.

```
root@localhost:~# passwd
New password:
Retype new password:
passwd: password updated successfully
root@localhost:~#
```

2. Create a repository file

http://classroom.example.com/content/rhel9.0/x86_64/dvd/AppStream

http://classroom.example.com/content/rhel9.0/x86_64/dvd/BaseOS

```
[local-BaseOS]
name=RHEL Local BaseOS
baseurl=http://classroom.example.com/content/rhel9.0/x86_64/dvd/BaseOS
enabled=1
gpgcheck=0

[local-AppStream]
name=RHEL Local AppStream
baseurl=http://classroom.example.com/content/rhel9.0/x86_64/dvd/AppStream
enabled=1
gpgcheck=0
```

3. Create a swap partition 512MB size.

```
root@servera:~# fdisk /dev/vdc
```

```
Welcome to fdisk (util-linux 2.40.2).
```

```
Changes will remain in memory only, until you decide to write them.  
Be careful before using the write command.
```

```
This disk is currently in use - repartitioning is probably a bad idea.  
It's recommended to umount all file systems, and swapoff all swap  
partitions on this disk.
```

```
Command (m for help): n
```

```
Partition type
```

```
  p   primary (1 primary, 0 extended, 3 free)  
  e   extended (container for logical partitions)
```

```
Select (default p):
```

```
Using default response p.
```

```
Partition number (2-4, default 2):
```

```
First sector (10487808-125829119, default 10487808):
```

```
Last sector, +/-sectors or +/-size{K,M,G,T,P} (10487808-125829119, default 125829119): +512M
```

```
Created a new partition 2 of type 'Linux' and of size 512 MiB.
```

```
Command (m for help): t
```

```
Partition number (1,2, default 2): 2
```

```
Hex code or alias (type L to list all): 82
```

```
Changed type of partition 'Linux' to 'Linux swap / Solaris'.
```

```
Command (m for help): w
```

```
The partition table has been altered.
```

```
Syncing disks.
```

```
root@servera:~# █
```

```
root@servera:~# mkswap /dev/vdc2
```

```
Setting up swapspace version 1, size = 512 MiB (536866816 bytes)
```

```
no label, UUID=3e25ce06-50d4-4267-9bde-53432b714014
```

```
root@servera:~# swapon /dev/vdc2
```

```
root@servera:~# echo "/dev/vdc1 swap swap defaults 0 0" >> /etc/fstab
```

```
root@servera:~# echo "/dev/vdc2 swap swap defaults 0 0" >> /etc/fstab
```

```
root@servera:~# swapon --show
```

NAME	TYPE	SIZE	USED	PRI0
/dev/dm-1	partition	3.9G	0B	-2
/dev/vdc1	partition	5G	0B	-3
/dev/vdc2	partition	512M	0B	-4

```
root@servera:~# █
```

4. Create one logical volume named database and it should be on datastore volume group with size 50 extent and assign the filesystem as ext3.

(i) the datastore volume group extend should be 8MiB. (ii) mount the logical volume under mount point /mnt/database.

5. Resize the logical volume size of 100 extent on /mnt/database directory.

6. Set the recommend tuned profile for your system.

```

root@servera:~# vgcreate -s 8M datastore /dev/vdc3
Physical volume "/dev/vdc3" successfully created.
Volume group "datastore" successfully created
root@servera:~# lvcreate -l 50 -n database datastore
Logical volume "database" created.
root@servera:~# mkfs.ext3 /dev/datastore/database
mke2fs 1.47.1 (20-May-2024)
Discarding device blocks: done
Creating filesystem with 409600 1k blocks and 102400 inodes
Filesystem UUID: 122f96ab-0081-49dc-a2d1-ba474baabd4d
Superblock backups stored on blocks:
    8193, 24577, 40961, 57345, 73729, 204801, 221185, 401409

Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done

root@servera:~# mkdir -p /mnt/database
root@servera:~# mount /dev/datastore/database /mnt/database
mount: (hint) your fstab has been modified, but systemd still uses
the old version; use 'systemctl daemon-reload' to reload.
root@servera:~# echo "/dev/datastore/database /mnt/database ext3 defaults 0 0" >> /etc/fstab
root@servera:~# lvextend -l 100 -r /dev/datastore/database
File system ext3 found on datastore/database mounted at /mnt/database.
Size of logical volume datastore/database changed from 400.00 MiB (50 extents) to 800.00 MiB (100 extents).
Extending file system ext3 to 800.00 MiB (838860800 bytes) on datastore/database...
resize2fs /dev/datastore/database
resize2fs 1.47.1 (20-May-2024)
Filesystem at /dev/datastore/database is mounted on /mnt/database; on-line resizing required
old_desc_blocks = 2, new_desc_blocks = 4
The filesystem on /dev/datastore/database is now 819200 (1k) blocks long.

resize2fs done
Extended file system ext3 on datastore/database.
Logical volume datastore/database successfully resized.

root@servera:~# tuned-adm profile $(tuned-adm recommend)
root@servera:~# tuned-adm active
Current active profile: virtual-guest
root@servera:~# vgs datastore
VG          #PV #LV #SN Attr   VSize  VFree
datastore   1   1   0 wz--n- 49.99g 49.21g
root@servera:~# df -hT /mnt/database
Filesystem                                Type      Size  Used Avail Use% Mounted on
/dev/mapper/datastore-database            ext3      740M   47K  704M   1% /mnt/database
root@servera:~# █

```